GLOBAL MENTAL HEALTH DEPRESSION ANALYSIS DASHBOARD

Prepared By

Soudeep Ghosh

Tools Used

- Power BI
- Power Query
- DAX
- Microsoft Excel
- GitHub

---

1. Introduction

Mental health disorders have become a major public health concern worldwide. Depression affects millions of people and often coexists with other mental health conditions such as anxiety, bipolar disorder, and substance-use disorders. Understanding the prevalence of these disorders and their associated factors is essential for improving healthcare policies and interventions.

This project presents an interactive Power BI dashboard designed to analyze global depression trends using multiple datasets. The dashboard explores various dimensions of depression, including age, gender, suicide rates, educational attainment, and the prevalence of other mental health disorders across countries.

2. Project Objectives

The objectives of this project were:

- To identify different types of mental health disorders represented in the dataset.
- To analyze depression prevalence across countries and over time.
- To determine whether depression is more prevalent among males or females.
- To identify the age groups most affected by depression.
- To explore the relationship between depression prevalence and suicide rates.
- To compare the prevalence of different mental health disorders.
- To assess the impact of educational attainment on depression rates.
- To identify regions with the highest burden of depression.
- To provide meaningful insights through interactive visualizations.

---

3. Datasets Used

The following datasets were used in this project:

- Depression prevalence by age group
- Depression prevalence among males and females
- Suicide rates versus depression prevalence
- Mental and substance-use disorder prevalence
- Depression by educational attainment
- Number of people suffering from depression

These datasets were cleaned and transformed using Power Query before being used in the dashboard.

4. Dashboard Overview

The Power BI dashboard consists of multiple analytical pages:

Executive Dashboard

Provides a high-level overview of depression prevalence, including global averages, geographical distribution, and trends over time.

Age Analysis

Examines depression prevalence across different age groups and identifies the most affected populations.

Gender Analysis

Compares depression prevalence between males and females across countries.

Suicide Analysis

Investigates the relationship between depression prevalence and suicide rates using scatter plot analysis.

Mental Disorder Analysis

Compares the prevalence of anxiety disorders, bipolar disorder, schizophrenia, eating disorders, drug-use disorders, alcohol-use disorders, and depression.

Education Analysis

Explores the relationship between educational attainment and depression prevalence among employed and active populations.

Additional Insights

Highlights the number of people suffering from depression and visualizes changes in depression prevalence over time.

Answer Page

Maps each project question to the corresponding dashboard page for easier interpretation and evaluation.

5. Methodology

The following steps were followed during the development of this project:

Data Cleaning

- Removed unnecessary records and invalid values.
- Eliminated BCE year entries and data inconsistencies.
- Corrected data types.
- Handled null values and transformation errors.

Data Transformation

- Applied filtering techniques.
- Promoted headers and formatted columns.
- Created calculated measures and aggregations.

Visualization Development

- Designed interactive dashboards using Power BI.
- Added slicers for country and year-based exploration.
- Incorporated maps, bar charts, line charts, scatter plots, and KPI cards.

---

6. Key Findings

The analysis revealed several important insights:

- Depression prevalence varies significantly across countries.
- Certain age groups exhibit higher depression prevalence than others.
- Depression prevalence differs between males and females.
- Countries with higher depression prevalence often demonstrate notable suicide burdens.
- Anxiety and depression are among the most prevalent mental health disorders.
- Educational attainment and employment status may influence depression outcomes.

- The education dataset contained information only for the year 2014, limiting longitudinal analysis in this area.
- Global mental health patterns are complex and influenced by multiple demographic and social factors.

---

7. Conclusion

This project demonstrates the use of Power BI to transform raw mental health datasets into meaningful and interactive visualizations. Through comprehensive analysis, the dashboard provides valuable insights into depression trends, demographic differences, and associated mental health factors.

The findings from this project emphasize the importance of data-driven approaches in understanding mental health challenges and supporting informed decision-making for healthcare planning and intervention strategies.

---

8. Future Scope

Future enhancements to this project may include:

- Incorporating additional population datasets to calculate depression burden relative to population size.
- Developing advanced DAX measures to identify countries with the fastest growth in depression prevalence.
- Integrating predictive analytics and machine learning models.
- Expanding the dashboard with real-time mental health indicators.

References

- Our World in Data
- Global Burden of Disease (GBD) Dataset
- Power BI Documentation
- Microsoft Power Query Documentation

Prepared by: Soudeep Ghosh

Project Title: Global Mental Health Depression Analysis Dashboard